Federico Leo Redi

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Academic Appointments

2026–Present **Associate Professor of Physics**, University of Bergamo and INFN Milan, Italy.
2023–2026 **Tenure-Track Assistant Professor of Physics**, University of Bergamo and INFN Milan, Italy.
2021–2023 **Senior Research Fellow**, Experimental Physics Department, CERN, Geneva, Switzerland.
2017–2021 **Postdoctoral Researcher**, High Energy Physics Laboratory, EPFL, Lausanne, Switzerland.

Education

2013–2017 **PhD in Fundamental Physics**, Imperial College London, United Kingdom. Thesis: *Searching for Heavy Neutral Leptons at LHCb and SHiP*. Supervisor: Prof. Andrey Golutvin.
2009–2013 **BSc and MSc in Physics, Distinction**, Imperial College London, United Kingdom.

Research

Experimental particle physics with LHCb (member since 2013; formerly SHiP and SND@LHC). Searches for physics beyond the Standard Model, in particular long-lived particles and heavy neutral leptons; precision electroweak measurements; trigger and detector development, including the VELO Upgrade II front-end ASIC and the GRENDEL detector proposal; analysis preservation and data reuse; machine learning for biomedical imaging.

Fellowships and Honours

2026 **National Scientific Habilitation for Full Professor** in Experimental Physics, Italy.
2022 **Ramón y Cajal Fellowship**, Agencia Estatal de Investigación, Spain (declined for the Bergamo position).
2021 **Beatriu de Pinós Fellowship**, Generalitat de Catalunya, Spain (declined for the CERN Fellowship).
2013–2017 **STFC PhD Studentship** and **Long-Term Attachments to CERN**, UK Science and Technology Facilities Council.

Leadership and Professional Service

2024–Present **Full Member, LHCb Editorial Board**; Technical Coordinator 2018–2024; Head of Institutional Reviewers since 2023.
2022–2024 **Convenor, QCD, Electroweak and Exotica Working Group**, LHCb; supervised approximately 30 analyses.
2021–2023 **Project Leader, Central Offline Selection (Stripping)**, LHCb; Co-coordinator, Offline Data Processing and Analysis; Member, Operations Planning Group.
2019–2022 **Co-convenor, LHC-wide Long-Lived Particles Working Group**; Convenor, LHCb Exotica and Higgs Physics (2019–2021).
2020–Present **Scientific Committee Member**, Workshop of the Long-Lived Particle Community.
2021–Present Grant reviewer for the Swiss, Spanish and Greek national funding agencies; referee for *Phys. Lett. B*, *Eur. Phys. J. C* and *SciPost Phys. Core*; Associate Editor, *Frontiers in Physics*.
2023–Present **University of Bergamo**: SUSTAIN PhD School Faculty Board, Selection Committee and Curriculum Coordinator (Electronics, Mechatronics and E-Mobility); Department Representative on the University Artificial Intelligence Working Group.

Selected Publications

[1] LHCb Collaboration, "Search for heavy neutral leptons in B-meson decays", *JHEP* **03** (2026) 178, arXiv:2512.14551.
[2] M. Citron, F. Redi and R. Schmitz, "Proposing GRENDEL: GalleRy ExperimeNt for Decays of Exotic LLPs at the LHC", arXiv:2609.00152 (2026).
[3] A. M. Abdullahi *et al.*, "The present and future status of heavy neutral leptons", *J. Phys. G* **50** (2023) 020501, arXiv:2203.08039.
[4] LHCb Collaboration, "Search for massive long-lived particles decaying semileptonically at $\sqrt{s} = 13$ TeV", *Eur. Phys. J. C* **82** (2022) 373, arXiv:2110.07293.
[5] LHCb Collaboration, "Measurement of the ratio of branching fractions $\mathcal{B}(\bar{B}^0 \to D^{*+}\tau^-\bar{\nu}_\tau)/\mathcal{B}(\bar{B}^0 \to D^{*+}\mu^-\bar{\nu}_\mu)$", *Phys. Rev. Lett.* **115** (2015) 111803, arXiv:1506.08614.

Selected Invited and Plenary Talks

2026 Neutrino Oscillation Workshop (NOW 2026), Otranto, Italy – parallel talk on heavy-neutral-lepton searches at the LHC.
2025 Lepton Photon 2025, Madison, United States – plenary talk on unusual signatures on behalf of ATLAS, CMS and LHCb.
2024 La Thuile 2024, Italy – plenary talk on electroweak physics and long-lived particles with LHCb.
2022 Lepton Photon 2022, Manchester, United Kingdom – invited plenary on long-lived particles for ATLAS, CMS and LHCb.

Teaching and Supervision

2022–Present **Course Convenor and Lecturer, Mechanics and Electromagnetism**, University of Bergamo; about 300 students per year.
2024–Present **Designer and Sole Lecturer, Physics of Climate Change**, University of Bergamo; approximately 50 students per year.
2018–2025 Co-supervised two PhD theses and supervised five MSc theses and five BSc projects (EPFL, CERN, Bergamo).

Outreach

2024–Present **LHCb Representative, IPPOG Steering Board**, CERN.
2015–Present **Head, LHCb Masterclass Programme**; Founding Convenor of the LHCb Outreach Working Group (2019–2021).